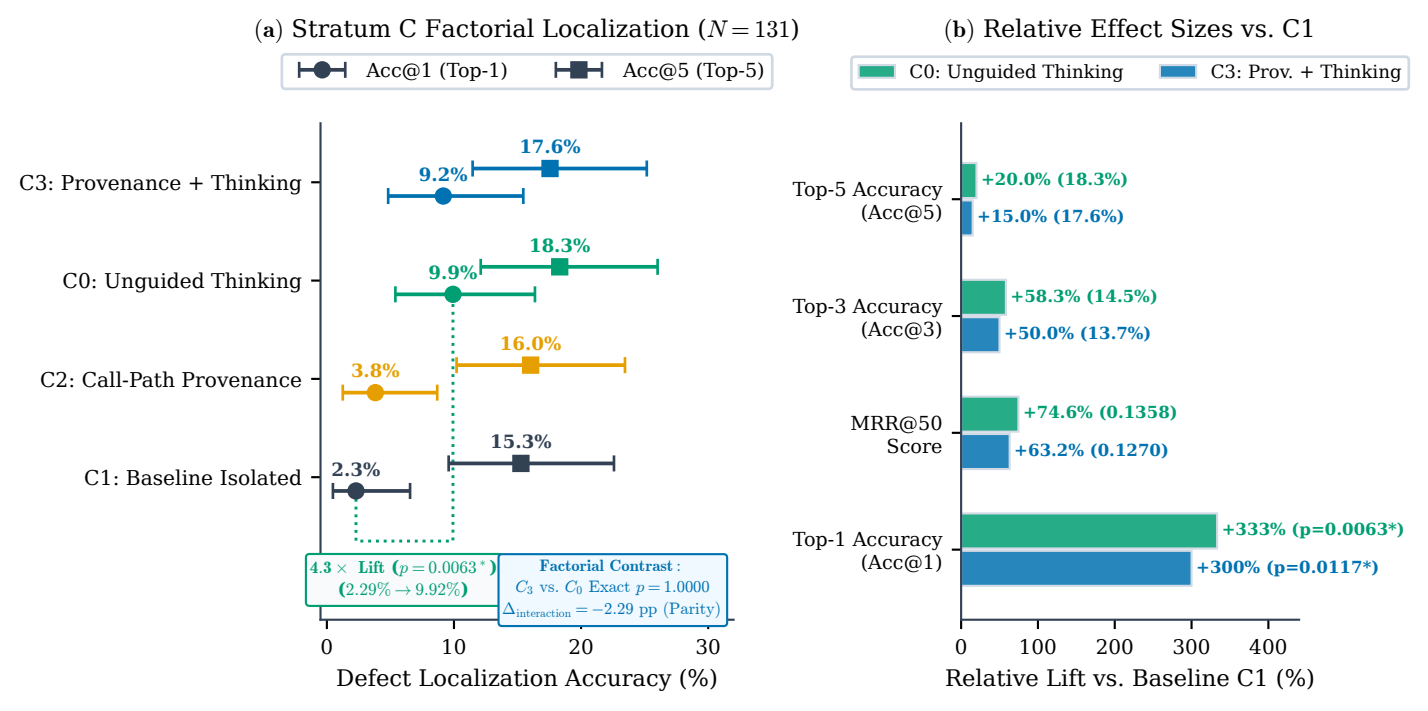
(a) Stratum C Factorial Localization ($N = 131$)
Acc@1 (Top-1)
Acc@5 (Top-5)
C3: Provenance + Thinking
C0: Unguided Thinking
C2: Call-Path Provenance
C1: Baseline Isolated
17.6%
9.2%
18.3%
9.9%
16.0%
3.8%
15.3%
2.3%
4.3× Lift ($p = 0.0063^*$)
(2.29% → 9.92%)
Factorial Contrast :
C_3 vs. C_0 Exact $p = 1.0000$
$\Delta_{\text{interaction}} = -2.29$ pp (Parity)
0
10
20
30
Defect Localization Accuracy (%)
(b) Relative Effect Sizes vs. C1
C0: Unguided Thinking
C3: Prov. + Thinking
Top-5 Accuracy (Acc@5)
+20.0% (18.3%)
+15.0% (17.6%)
Top-3 Accuracy (Acc@3)
+58.3% (14.5%)
+50.0% (13.7%)
MRR@50 Score
+74.6% (0.1358)
+63.2% (0.1270)
Top-1 Accuracy (Acc@1)
+333% (p=0.0063*)
+300% (p=0.0117*)
0
100
200
300
400
Relative Lift vs. Baseline C1 (%)